MD. KHAIRUL BASHAR BHUIYAN

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Google Scholar ResearchGate GitHub LinkedIn

RESEARCH INTERESTS

Machine Learning, Data Science, Cloud Computing, Quantum Machine Learning, IoT

RESEARCH SUMMARY

Efficiency-focused ML and QML researcher aiming towards developing lightweight, explainable and robust Machine Learning models, applicable across multiple real-world domains. Track record includes 7 peer-reviewed publications with 100 citations, QC Boot-Camp, IEEE IEACon oral+poster presenter with 1.5+ years of RA and local industry-grade job experience.

EDUCATION

Royal University of Dhaka(RUD), Dhaka, Bangladesh — September, 2024 — August, 2025
Master of Science in Computer Science. — Cumulative GPA: 3.53/4.00

- **Thesis Title:** A Mathematical Framework for Compliance Assessment: An In-Depth Insights into Bangladesh's Ready-Made Garments Industry

BRAC University, Dhaka, Bangladesh — January, 2016 — April, 2021
Bachelor of Science in Electrical and Electronic Engineering - Major in Electronics. — Cumulative GPA: 2.78/4.00

- **Thesis Title:** Modeling and Simulation of a Satellite Training Kit for Providing Space Systems Engineering Knowledge and Technological Development in STEAM

RESEARCH EXPERIENCE

Mahdy Research Academy — Website — Dhaka, Bangladesh
North South University
ML/QML Researcher(Remote) — January, 2024 — December, 2024

- Achieved hands-on knowledge about Quantum Mechanics, QML, Quantum Security, Key Distribution Protocols, Quantum Error Correction, Fault Tolerance, Quantum Energy Teleportation, QISKIT Coding, Novel Research Ideas, Conclusive Research.
- Conducted research regarding a hybrid Classical-Quantum Machine Learning Framework for skin cancer detection.

Control Application and Research Center(CARC) — Website — Dhaka, Bangladesh
BRAC University
Undergraduate Research Assistant — September, 2021 — July, 2022

- Helped the professor supervising the undergraduate thesis groups.
- Conducted research regarding renewable energy and control system application.

PROJECTS

Artificially Intelligent System to Pass on Unbiased Judgements to Criminal Cases — August, 2019

- The purpose of the developed system was to assist the judicial system. The base model was constructed to detect changes in body resistance, heart rate, and facial expressions. First, we constructed a simple lie detector circuit for measuring the body resistance. This lie detector circuit uses our body resistance to glow an LED bulb. Now, the body's resistance would be low due to excessive sweating. As a result, the LED would glow very dimly. On the other hand, the LED would glow brightly as the body resistance is high, which indicates the user is not stressed. Next, we used an ECG sensor embedded with an Arduino UNO processor to measure the change in heart rate. If the user plugged in is under stress, there would be a sudden high pulse on the serial plotter. If not, then the pulses would go normally. For the final demonstration, we use a data-based facial recognition system. This machine-learning model allows us to read the facial gesture-posture changes during interrogation. It can detect some moods such as "happy", "disgusted", "angry" and "neutral". Based on our data, we determine "anger" and "disgust" as signs of guilt and the others as normal signs. Now, if we combine these three readings, then we can use this as a simple polygraph system. By this system, we can detect whether the user is liable for the accusation or not.

Satellite Training Kit May, 2021

- This project can help the students to learn about different subsystems of a basic satellite, how to assemble the different sub-systems into a fully functional system and how to test or debug it once it is incorporated. The Satellite Training Kit is designed to help students understand satellite communication through practical experiments.
  — GitHub

Traffic-safety Project August, 2024

- This project is about providing traffic safety. The trained model uses Mosaic Dataloader to improve detection accuracy. It detects motorcyclists, their helmets and license plates in diverse, cluttered urban environments. Each label includes a probability score. The bounding boxes are well-aligned with the actual objects, showing good spatial precision and scalable surveillance. It can help identify riders without helmets, detect license plates for registration checks and vehicle-tracking for law enforcement.
  — GitHub

PUBLICATIONS

Google Scholar statistics: Total of 101 citations as of May 15, 2026, with h-index = 2, i10-index = 2
Researchgate statistics: Total of 89 citations as of May 15, 2026, with h-index = 3, Research Interest Score = 67.4

Journal paper

SG-PBFS: Shortest Gap-Priority Based Fair Scheduling technique for job scheduling in cloud environment

- Saydul Akbar Murad, Zafril Rizal M Azmi, Abu Jafar Md Muzahid, **Md. Khairul Bashar Bhuiyan**, Md Saib, Nick Rahimi, Nusrat Jahan Prottasha, Anupam Kumar Bairagi — ELSEVIER

Priority based job scheduling technique that utilizes gaps to increase the efficiency of job distribution in cloud computing

- Saydul Akbar Murad, Zafril Rizal M Azmi, Abu Jafar Md Muzahid, Md Murad Hossain Sarker, M Saef Ullah Miah, **Md. Khairul Bashar Bhuiyan**, Nick Rahimi, Anupam Kumar Bairagi — ELSEVIER

Neural Network-based Study for Rice Leaf Disease Recognition and Classification: A Comparative Analysis Between Feature-Based Model and Direct Imaging Model

- Farida Siddiqi Prity, Mirza Raquib, Saydul Akbar Murad, Md Jubayar Alam Rafi, **Md. Khairul Bashar Bhuiyan**, Anupam Kumar Bairagi —WILEY

MRAN-VQA: Multimodal Recursive Attention Network for Visual Question Answering

- Mohammad Shariful Islam, Mohammad Abu Tareq Rony, Md Murad Hossain Sarker, **Md. Khairul Bashar Bhuiyan**, Md Saib, Md Aktarujjaman, Md Shahab Uddin, Abeer D Algarni, Ahmad Taher Azar, Walid El-Shafai — ELSEVIER

Revolutionizing Cloud Computing: A Novel Approach with Shortest Gap Priority-Based Fair Scheduling (SG-PBFS) for Optimal Job Scheduling

- Saydul Akbar Murad, Zafril Rizal M Azmi, Md Murad Hossain Sarker, Mohammad Kamrul Hasan, **Md. Khairul Bashar Bhuiyan**, Md Saib, Anupam Kumar Bairagi—Submitted to **Computer Communications** Journal

A Mathematical Framework for Compliance Assessment: An In-Depth Insights into Bangladesh's Ready-Made Garments Industry

- Md Murad Hossain Sarker, Md Saib, **Md. Khairul Bashar Bhuiyan**, Md. Ridwan Ali, Naznin Nahar — Pre-print

A Hybrid Classical-Quantum Machine Learning Framework for Skin Cancer Detection Using A Knowledge Distillation Strategy

- Md Murad Hossain Sarker, **Md. Khairul Bashar Bhuiyan**, Md Saib, Mohammad Junayed Hasan, M.R.C. Mahdy —Pre-print

Conference paper

MemeFusionNet: A Cross-Linguistic Multimodal Model for Identifying Troll Memes

- Tipu Sultan, Hadi Ali Akbarpour, Walid El-Shafai, Md Saib, **Md. Khairul Bashar Bhuiyan**, Md Shariful Islam, Ahmad Taher Azar, Chakib Ben Njima —IEEE

ViTDeBERTaNews: A Comparative Study of Single-modal, Multimodal, and LLM Techniques for Detecting Fake News

- Tipu Sultan, Md Murad Hossain Sarker, **Md. Khairul Bashar Bhuiyan**, Md Saib, Mohammad Shariful Islam, MD. Rasel Hossain, Walid El-Shafai, Ahmad Taher Azar, Chakib Ben Njima —IEEE

Modeling and Simulation of a Satellite Training Kit for Providing Space Systems Engineering Knowledge and Technological Development in STEAM

- **Md Khairul Bashar Bhuiyan**, Md Shoaib Islam, Tanjidul Islam Sanvi, MD Jahid Hashan Talukder, Raihana Shams Islam Antara, AKM Abdul Malek Azad — IEEE

SELECTED COURSES

Master's Notable Courses

- MS 661 - Advanced Artificial Intelligence
- MS 641 - Software Engineering and Project Management
- MS 652 - Software Quality Assurance
- MS 601 - Distributed Database Query Optimization
- MS 691 - Data Mining and Warehousing
- MS 671 - Object Oriented Programming Language

Bachelor's Notable Courses

- EEE 305 - Control Systems
- EEE 307 - Opto-Electronic Devices
- EEE 365 - Microprocessors
- EEE 343 - Digital Signal Processing
- EEE 415 - Analog Integrated Circuit Design
- EEE 413 - Digital System Design
- EEE 411 - VLSI Design

Additional Online Courses

Advanced Machine Learning and Deep Learning Course by Shakhawat Hossain September, 2022 — April, 2024

- Ensemble Learning, Cross Validation, DBSCAN, AHP, TOPSIS, WASPAS, Prophet algorithm, Mean Shift, Birch Clustering, Dimensionality Reduction, CNN, Reinforcement Learning, 30 Points of Regression, Time Series Analysis, Feature Engineering, Neural Network, ANN, RNN, Natural Language Processing, MLOps.

Supervised Remote Sensing Course by Dataskool February, 2023 — June, 2023

- Classification, Mapping, Superimpose, Datum, Spatial Data, Map Projection, Raster Analysis, Interpolation, Satellite Image Processing, GIS.

Introduction to Quantum Computing by The Coding School(TCS) September, 2024 — April, 2025

- Quantum Gates, Quantum Entanglement, Quantum Error Correction, Quantum Networking, Quantum Algorithms and protocols, Quantum Hardware, Programming with Cirq, Superconducting Qubits.

Quantum Computing Research Course by Mahdy Research Academy January, 2024 — December, 2024

- Quantum Mechanics, Quantum Machine Learning, Quantum Security, Key Distribution Protocols, Quantum Error Correction, Fault Tolerance, Quantum Energy Teleportation, QISKIT Coding, Novel Research Ideas, Conclusive Research.

Quantum Computing Boot-camp by Quantum School Bangladesh April, 2025 — August, 2025

- Fundamentals of Quantum Information, Computation, Cryptography, Secure Communication, Error and Noise, Mathematical and Theoretical Foundation

The Ultimate Hands-On Hadoop: Tame your Big Data! by Udemy August, 2025 — September, 2025

- HDFS Architecture, Map Reduce, Non-Relational Databases, YARN, Zookeeper, Apache Spark, Kafka, Sqoop, Pig, Hive, MySQL, Drill, Flink.

Cloudera Hadoop Administration by class central September, 2025 — October, 2025

- CDH Cluster, Amazon AWS, LDAP, Kerberos, Hadoop, Cassandra, YARN, Hive, Metastore.

Confluent Apache Kafka Fundamentals Accreditation by Confluent November, 2025 — Present

- Confluent Kafka servers, Broker, Producers, Consumers, Partition, Key, Value, Timestamp, KRaft, Zookeeper.

AWARDS

Merit-based Board-Scholarship Cumilla Education Board
Stipend based on Secondary School Certification Examination. Grade - Talentpool May, 2012

Merit-based Board-Scholarship Dhaka Education Board
Stipend based on Higher Secondary School Certification Examination. Grade - General. August, 2014

FFQ Merit-based Scholarship BRAC University
Stipend based on Academic excellence. May, 2016 — December, 2018

Presentation Skill Certification Residential Campus, BRAC University, Dhaka
Based on Residential Semester Activity for Presentation Skill. December, 2016

JOB EXPERIENCES

Brain System — 🌐 Website Dhaka, Bangladesh
Internship August, 2023 — September, 2024

- Data Structure and Data Pre-processing
- Machine Learning Applications

Fulltime, Machine Learning Engineer April, 2026 — Present

- ERP Implementation and Custom AI development
- Big-Data and Data Mining and LLM model development

Zylo — 🌐 Website Dhaka, Bangladesh
Fulltime, Big-Data Engineer September, 2024 — December, 2025

- Data Visualization, Data Pipe-line and Warehousing
- Data Engineering: ETL, Streaming
- Data Strategy and Architecture

ENGLISH Proficiency TEST

IELTS (Academic): 6.5 (overall score)
Listening: 6.5 — Reading: 7.0 Speaking: 6.5 — Writing: 6.0
Test date: 16 December 2022

SKILLS

- **Programming:** Java (Basic), C (Basic), Python (Intermediate)
- **Database:** MySQL, MongoDB, NoSQL
- **Software:** SolidWorks, MATLAB Simulink, IBM SPSS, IBM QISKIT, ArcGIS, Cloudera Manager, HDFS Architecture, Starburst, Apache Kafka, Confluent Kafka, Hive, Spark, LaTex
- **IDE:** Arduino, Proteus, PyCharm, Jupyter Notebook, Google Colab
- **Library:** Numpy, Pandas, Matplotlib, Scikit-learn, TensorFlow, PyTourch, Django
- **Soft Skills:** Presentation Skill, Team Work, Adaptability, Communication Skill

REFERENCES

Dr. AKM Abdul Malek Azad
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Scholar Profiles: A.K.M Abdul Malek Azad — Google Scholar — Researchgate

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